Basic integration formulas

1. $\int k\ dx = kx + C$ (any number k)
2. $\int x^n\ dx = \frac{x^{n+1}}{n+1} + C \qquad (n \neq -1)$
3. $\int \frac{dx}{x} = \ln|x| + C$
4. $\int e^x\ dx = e^x + C$
5. $\int a^x\ dx = \frac{a^x}{\ln a} + C \qquad (a > 0,\ a \neq 1)$
6. $\int \sin x\ dx = -\cos x + C$
7. $\int \cos x\ dx = \sin x + C$
8. $\int \sec^2 x\ dx = \tan x + C$
9. $\int \csc^2 x\ dx = -\cot x + C$
10. $\int \sec x \tan x\ dx = \sec x + C$
11. $\int \csc x \cot x\ dx = -\csc x + C$
12. $\int \tan x\ dx = \ln|\sec x| + C$
13. $\int \cot x\ dx = \ln|\sin x| + C$
14. $\int \sec x\ dx = \ln|\sec x + \tan x| + C$
15. $\int \csc x\ dx = -\ln|\csc x + \cot x| + C$
16. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$
17. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$

Formulas in Chapter 6

1. **(a) Volume by Disks for Rotation About the x-Axis**

$$V = \int_a^b A(x)dx = \int_a^b \pi[R(x)]^2 dx$$

   (b) Volume by Washers for Rotation About the x-Axis

$$V = \int_a^b A(x)dx = \int_a^b \pi\left([R(x)]^2 - [r(x)]^2\right)dx$$

2. **Shell Formula for Revolution About a Vertical Line**

$$V = \int_a^b 2\pi \begin{pmatrix}\text{shell}\\ \text{radius}\end{pmatrix}\begin{pmatrix}\text{shell}\\ \text{height}\end{pmatrix} dx = \int_a^b 2\pi R(x)h(x)dx$$

Special case: Revolution of the region between $y = f(x)$ and $y = 0$ about the y-axis

$$V = \int_a^b 2\pi R(x)h(x)dx = \int_a^b 2\pi x f(x)dx$$

3. **The Length of the Curve** $y = f(x)$

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx$$

4. **Area of the Surface Generated by Revolution of** $y = f(x)$ **About the** x**-Axis**

$$S = \int_a^b 2\pi y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx = \int_a^b 2\pi f(x)\sqrt{1 + [f'(x)]^2}\,dx$$